

ITEC 3100 – Introduction to Networks

Professor: Dr. Binh Tran

Lab 4b:

Please use a different color font for your answers so I can clearly see them:

Q1. Fill in the blanks in the table regarding Wifi:

Wifi Standard	Maximum Speed	Frequency
802.11a	54 Mbps	5 GHz
802.11b	11 Mbps	2.4 GHz
802.11g	54 Mbps	2.4 GHz
802.11n	600 Mbps	2.4/5 GHz
802.11ac	1.3 Gbps	5 GHz

Q2. List 3 benefits of using Wifi?

Easy setup, Convenience, Multiple devices

Q3. List 3 disadvantages of Wifi?

Can be intercepted, Interference, not as fast as ethernet

Q4. Locate your home wireless router insert picture and researching the model number identify which 802.11 standard it is: (a,b,g,n,ac)? Is it single, dual or tri-band?

NETGEAR Nighthawk Smart Wi-Fi Router (R6900P) - AC1900

802.11ac, Dual band



Q5. Describe how we use all 3 types of cables when connecting a wired laptop to the Internet.

Fiber Optic connects to an Optical Network Terminal

Coaxial is the main cable for modem

Twisted Pair plugs one end into LAN port of router and the other end to the RJ45 port.

Watch video on Network tools:

Networking Tools:

<https://www.youtube.com/watch?v=HAceoquch1c>

Q6. Which tool is used to send a signal and receive it to make sure the data is accurate through a CAT cable?

Time Domain Reflectometer

Q7. Which tool is used to locate where a port is connected to?

Tone Generator

Q8. Which tool is used to confirm the speed and that cable is properly wired?

Cable Tester

Q9. Which tool is used to send a signal and receive it to make sure the data is accurate through a Fiber optic cable?

Optical Time Domain Reflectometer

Watch video on Wifi Security:

Wifi Security:

<https://www.youtube.com/watch?v=WZalfyvERcA>

Q10. Which security standard should we use today and which one should we avoid?

Use WPA2 and avoid WEP and WPA

Q11. Insert screenshots we took in class

LINKSYS
A Division of Cisco Systems, Inc.

Firmware Version: 1.52.02

Wireless-B Broadband Router BEFW1154

Wireless

Setup | **Wireless** | Security | Applications & Gaming | Administration | Status

Basic Wireless Settings | **Wireless Security** | Wireless Network Access | Advanced Wireless Settings

Wireless Security

Wireless Security: ☒ Enable ☐ Disable

Security Mode: **WPA Pre-Shared Key**

WPA Algorithm: **TKIP**

WPA Shared Key:

Group Key Renewal: seconds

Wireless Security

The router supports four different types of security settings for your network: WPA Protected Access (WPA) Pre-Shared Key, WPA Remote Access Dial In User Service (RADIUS), RADIUS, and WPA Equivalent Protection (WEP).

[More...](#)

Save Settings **Cancel Changes**

CISCO SYSTEMS

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Wireless

Setup | **Wireless** | Security | Applications & Gaming | Administration | Status

Basic Wireless Settings | **Wireless Security** | Wireless Network Access | Advanced Wireless Settings

Wireless Network

Wireless: ☒ Enabled ☐ Disabled

Wireless Network Name(SSID):

Wireless Channel: **6 - 2.442 GHZ**

Wireless SSID Broadcast: ☐ Enabled ☒ Disabled

Basic Wireless Settings

The Wireless screen allows you to customize data transmission settings. In most cases, the advanced settings on this screen should remain at their default values.

[More...](#)

Save Settings **Cancel Changes**

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Wireless-B Broadband Router BEFW1154

Setup

Setup | Wireless | Security | Applications & Gaming | Administration | Status

Basic Setup | DDNS | MAC Address Clone | Advanced Routing

Internet Setup

Internet Connection Type

Optional Settings (required by some ISPs)

Obtain an IP automatically

Host Name:

Domain Name:

MTU: ☐ Enable ☒ Disable Size:

Speed & Duplex: ☐ 100 Mb Full ☒

Local IP Address: . . .

Subnet Mask: . . .

Local DHCP Server: ☒ Enable ☐ Disable

Start IP Address:

Number of Address:

DHCP Address Range: -

Client Lease Time: minutes (0 means two day)

Static DNS 1: . . .

Static DNS 2: . . .

Static DNS 3: . . .

WINS: . . .

Basic Setup

The Basic Setup screen is where basic configuration is performed. Some ISPs (Internet Service Providers) will require that you enter the DNS information. These settings can be obtained from your ISP. After you have configured these settings, you should set a router password from the Administration->Management screen.

Completing the Internet Setup section is all that is required to set up for your specific ISP. Please look at the table below to configure the Router for your Internet connection.

[More...](#)

Save Settings **Cancel Changes**

CISCO SYSTEMS



Complete and upload to the Lab 4 dropbox.